Unit 5 - Ratios and Rates

The following B.E.S.T. Standards will be covered in this unit:

MA.6.AR.3.1 Given a real-world context, write and interpret ratios to show the relative sizes of two quantities using appropriate notation: $\frac{a}{b}$, a to b, or $a{:}b$, where $b \neq 0$.

- *Clarification 1: Instruction focuses on the understanding that a ratio can be described as a comparison of two quantities in either the same or different units.*
- *Clarification 2: Instruction includes using manipulatives, drawings, models, and words to interpret part-to-part ratios and part-to-whole ratios.*
- *Clarification 3: The values of a and b are limited to whole numbers.*

MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.

- *Clarification 1: Instruction includes using manipulatives, drawings, models, and words and making connections between ratios, rates, and unit rates.*
- *Clarification 2: Problems will not include conversions between customary and metric systems.*

MA.6.AR.3.3 Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios.

- *Clarification 1: Instruction includes using two-column tables (e.g., a relationship between two variables) and three-column tables (e.g., part-to-part-to-whole relationships) to generate conversion charts and mixture charts.*

MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.

Clarification 1: Instruction includes the comparison of $\frac{part}{whole}$ to $\frac{percent}{100}$ in order to determine the percent, the part, or the whole.

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

- *Clarification 1: Instruction includes the use of tables, tape diagrams, and number lines.*

Unit 5, Lesson 1: What Are Ratios?

Benchmark

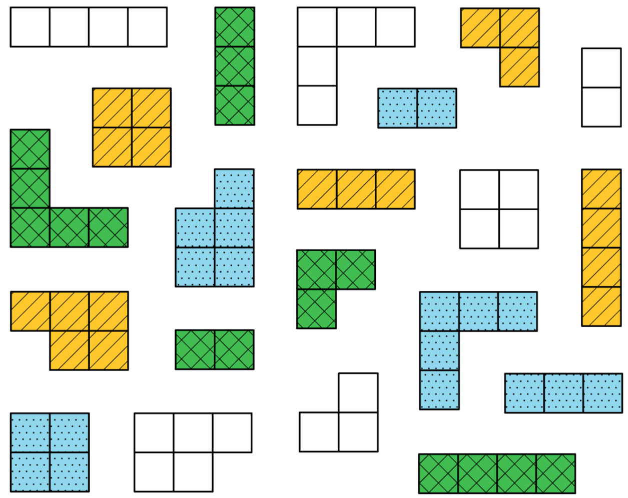
MA.6.AR.3.1 Given a real-world context, write and interpret ratios to show the relative sizes of two quantities using appropriate notation: $\frac{a}{b}$, a to b, or $a:b$ where $b \neq 0$.

Learning Targets

- ☐ I can write and interpret ratios.
- ☐ I can draw a diagram that represents a ratio and explain what the diagram means.

5.1.1 Warm-Up: What Kind and How Many?

A set of figures with different colors, shapes, and sizes is shown.



The figures can be sorted using different categories.

1. Complete the table by determining the number of groups for each method of sorting. For the last row, determine a third way the figures could be sorted and how many groups the sorting method would create.

Sorting Method	Number of Groups Created
Sorted by color	
Sorted by area	
Sorted by ______________	

5.1.2 Guided Instruction: Writing and Interpreting Ratios

The figures from the Warm-Up can be described using ratio language.

A ***ratio*** is a comparison of two or more quantities.

1. Complete the statements to describe the relationships between the figures from the Warm-Up.

    a. There are _____ yellow figures for every _____ green figures.

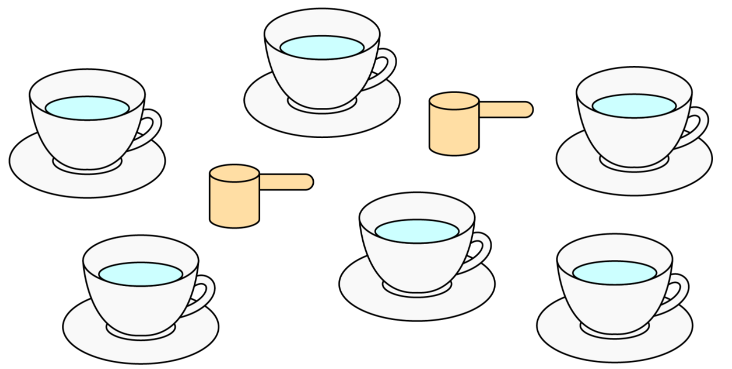
    b. For every _____ figures with an area of 2 square units, there are _____ figures with an area of 5 square units.

    c. The ratio of square-shaped figures to L-shaped figures is _____ to _____.

Ratios can be written in three ways. The ratio of blue figures to white figures is expressed in all three ways below.

Method for Writing Ratios	The Ratio of Blue Figures to White Figures
Using a colon (:)	$5 : 6$
Using the word "to"	5 to 6
Using a fraction bar	$\frac{5}{6}$

2. Avril makes a tea called chai using a pre-made powdered mix. The box instructions say to mix 2 scoops of chai powder with 6 cups of hot water.



Complete each description of the collection by writing a number in each box.

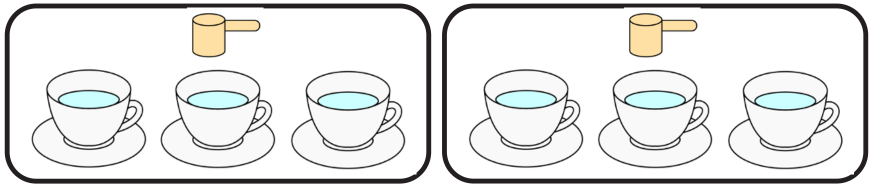
a. The ratio of chai mix scoops to cups of water is

$\boxed{} : \boxed{}$.

b. The ratio of cups of water to chai mix scoops is

$\frac{\boxed{}}{\boxed{}}$.

3. The relationship between chai mix and water can also be described using other numbers. Consider the rearranged image.



Complete the statements.

a. There are _____ cups of water per scoop of chai mix.

b. There is _____ scoop of chai mix for every cup of water.

The order of the ratio is important, as it relates specifically to the quantities being described. The order used in the written description must match the order of the values that are being compared.

4. Discuss with your partner why the ratio of chai mix scoops to cups of water is not equivalent to the ratio of cups of water to chai mix scoops.

5.1.3 Try It: Writing and Interpreting Ratios

In a fruit bowl, there are 6 oranges, 4 apples, and 2 plums.

1. Complete the statements to describe the contents of the fruit bowl.

a. The ratio of oranges to apples is _____ : _____.

b. The ratio of plums to apples is _____ to _____.

c. For every _____ apples, there are _____ plums.

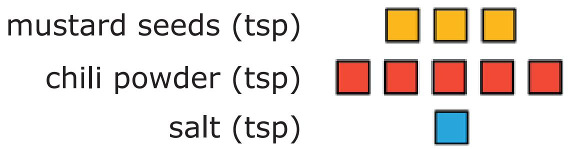
d. For every 3 oranges, there is one
○ apple.
○ plum.

2. Explain how you determined the answer for Part d using words or a sketch.

5.1.4 Guided Instruction: Representing Ratios With Diagrams

When solving problems involving ratios, it can be helpful to use diagrams and number lines to represent the relationships described.

1. A recipe for 1 batch of spice mix says, "Combine 3 teaspoons (tsp) of mustard seeds, 5 teaspoons of chili powder, and 1 teaspoon of salt." Below is a diagram to represent this situation.



   How many teaspoons of spices are in 1 batch of the mix?

Ratios can be used to describe part-to-part relationships, where one part is being compared to another, and part-to-whole relationships, where one part is being compared to the entire group.

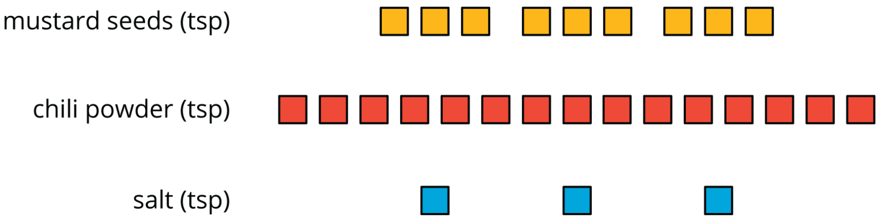
2. Complete the statements in the table.

Type of Relationship	Example Statements
part-to-part	The ratio of tsp of chili powder to tsp of mustard seeds is ____:____. There are _____ teaspoons of mustard seeds per tsp of salt.
part-to-whole	The ratio of teaspoons of chili powder to teaspoons of spice mix is _____ to _____. For every _____ teaspoon(s) of salt, _____ teaspoon(s) of spice mix can be made.

3. Discuss with your partner what is meant by "whole" in part-to-whole.

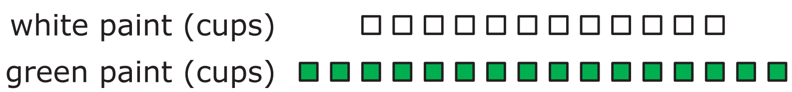
5.1.5 Collaboration: Interpreting Ratios With Diagrams

1. The diagram shows the ingredients Luca mixed to make the same spice mix recipe.



Work with your partner to determine how many batches of spice mix are represented by the diagram. Explain or show your reasoning.

2. When mixing paint colors, the same shade is produced when the same ratios of each color are used to mix the paint. The diagram represents 4 batches of light green paint that Mabelise mixed.



Use the diagram to answer the following questions with your partner.

a. Draw a diagram that represents 1 batch of the same shade of light green paint.

b. Complete the statements to describe the relationships.

- For every _____ cups of white paint, there are _____ cups of green paint in the mix.
- The ratio of [○ white paint ○ green paint] to [○ white paint ○ green paint] is $\frac{4}{3}$.
- The ratio of white paint to the light green mixture is _____ to _____.
- There are _____ cups of green paint used for every 14 cups of light green paint that is mixed.

5.1.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current level of understanding is related to each target.

1. I can write and interpret ratios.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

2. I can draw a diagram that represents a ratio and explain what a diagram means.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

Unit 5, Lesson 2: Using Models to Represent Ratios

Benchmark

MA.6.AR.3.1 Given a real-world context, write and interpret ratios to show the relative sizes of two quantities using appropriate notation: $\frac{a}{b}$, a to b, or $a{:}b$ where $b \neq 0$.

Learning Targets

- ☐ I can write and interpret ratios.
- ☐ I can represent ratios using various models.

5.2.1 Warm-Up: Nana's Chocolate Milk

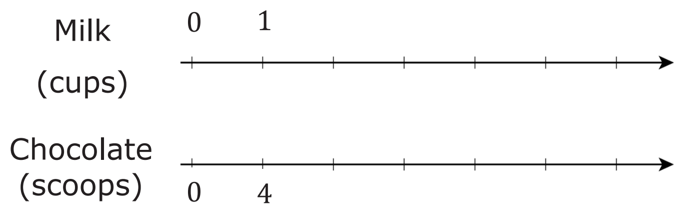
1. Dan's Nana prefers her chocolate milk made with just the right "chocolateyness," but he made a mistake when mixing it for her. The image below shows what Dan should have mixed and what he actually mixed.

   a. Describe what Dan did wrong.

   b. How could Dan fix it so that the glass contains Nana's preferred balance of milk and chocolate?

5.2.2 Exploration: Using Double Line Diagrams to Represent Ratios

Nana's chocolate milk recipe calls for 4 scoops of chocolate for every 1 cup of milk. One way to represent the recipe is with a double number line diagram.



1. Label each tick mark on the double number line diagram to represent the relationship between cups of milk and chocolate scoops.

2. How does the double number line show the ratio of Nana's preferred balance of milk and chocolate?

3. How many scoops of chocolate need to be mixed with 5 cups of milk to maintain the same ratio of milk to chocolate from Nana's recipe?

4. Circle where the answer to the previous question is represented on the double number line diagram.

Each pair of numbers on the double number line diagram represents an equivalent ratio. For example, the ratio $6 : 24$ is equivalent to the ratio $2 : 8$.

5. Complete the statement to interpret these two equivalent ratios in the context of Nana's recipe.

   If 6 cups of milk are mixed with _____ scoops of chocolate, it will taste the same as mixing _____ cups of milk with 8 scoops of chocolate.

6. Recall Dan's error when mixing Nana's chocolate milk.

Nana's Recipe	What Dan Accidentally Mixed
1 cup milk 4 scoops chocolate	1 cup milk 5 scoops chocolate

   a. Sabrina said Dan's mistake can be fixed by adding 0.5 cup of milk and 1 scoop of chocolate to the glass. Discuss with your partner whether or not you think Sabrina is correct.

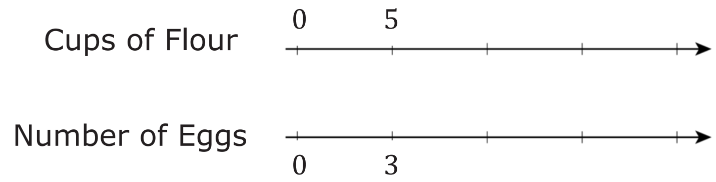
   b. Complete the statement.

   If Dan follows Sabrina's recommendation, the glass will contain _____ cups of milk and _____ scoops of chocolate.

7. Use the double number line diagram to determine whether Sabrina's recommendation will result in chocolate milk with Nana's preferred taste.
   If Sabrina's recommendation will work, mark the locations of this mixture on the double number line diagram with two Xs, one on each number line.

5.2.3 Try It: Using Double Line Diagrams to Represent and Interpret Ratios

1. A recipe for 1 batch of cookies calls for 5 cups of flour and 3 eggs. A double line diagram is shown to represent the recipe.



   a. What is the ratio of flour to eggs in 1 batch of cookies?

   b. Complete the diagram to show the amount of flour and eggs needed for 2, 3, and 4 batches of cookies.

   c. How many eggs and cups of flour are used in 4 batches of cookies?

   d. Complete the ratio comparing cups of flour to 6 eggs.

   $$\boxed{} : 6$$

   e. Complete the ratio of eggs to 15 cups of flour.

   $$\frac{\boxed{}}{15}$$

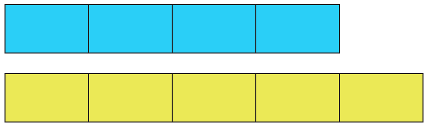
5.2.4 Guided Instruction: Using Tape Diagrams to Represent Ratios

A tape diagram is another way to represent ratios. All the parts of the diagram that are the same size have the same value, similar to increments on a number line.

The tape diagram shown represents the ratio of cats to dogs at a local animal shelter.

Number of Cats

Number of Dogs



1. Complete the statements using the tape diagram.

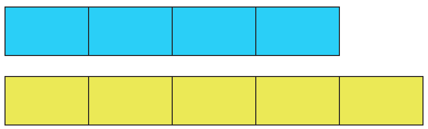
   a. The ratio of dogs to cats is $\frac{\square}{\square}$.

   b. For every _____ cats at the shelter, there are _____ dogs.

2. If there are 27 cats and dogs at the animal shelter in all, and the ratio of cats to dogs is $4\!:\!5$, the same tape diagram can be used to represent the relationship.

Number of Cats

Number of Dogs



   a. What is the value of each small rectangle in the diagram given this new information?

   b. Label each small rectangle with this value.

   c. How many dogs are at the animal shelter?

   d. How many cats are at the animal shelter?

   e. Complete the statement.

   The ratio of cats to dogs at the shelter, $4\!:\!5$, is equivalent to the ratio _____ to _____.

5.2.5 Your Turn!

1. Mr. Floyd is planning a field trip for his class to the Cade Museum for Creativity and Invention in Gainesville, Florida. To meet the museum's recommended ratio of chaperones to students, Mr. Floyd needs to have 1 adult chaperone for every 9 students. The tape diagram represents the ratio of adults to students needed.

Number of Adults

Number of Students

   a. What is the ratio of students to chaperones required by the museum?

   There are 72 students who will attend the trip from Mr. Floyd's classes. Use the tape diagram to determine how many chaperones will be needed.

   b. Label each small rectangle with the number of people it represents.

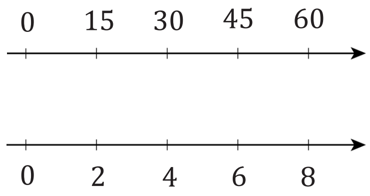
   c. Complete the statement.

   For all 72 students to attend, _____ chaperones are needed.

2. A cashier worked an 8-hour day and earned $\$60.00$. The double number line represents the amount she earned for working different numbers of hours.

Wages Earned (dollars)

Time Worked (hours)



   a. Determine the amount of money she earned in 1 hour. Show your work and label it on the double number line.

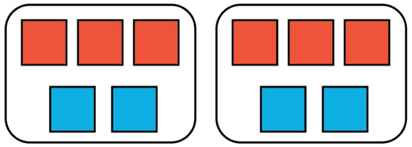
   b. How much money does the cashier earn if she works 3 hours?

5.2.6 Wrap-Up: Ticket Out the Door

1. The representation shows the amount of red and blue paint that makes 2 batches of a particular shade of purple paint called lilac.

Cups of Red Paint

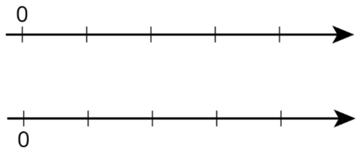
Cups of Blue Paint



   a. On the double number line, label the tick marks to represent amounts of red and blue paint used to make batches of this shade of purple paint.

Cups of Red Paint

Cups of Blue Paint



   b. How many batches are made with 12 cups of red paint?

   c. How much red paint should be mixed with 6 cups of blue paint to make lilac paint?

   d. Complete the statement.

   To make this shade of lilac paint, mix _____ cups of blue paint for every _____ cups of red.

Unit 5, Lesson 3: Exploring Rates

Benchmark

MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.

Learning Targets

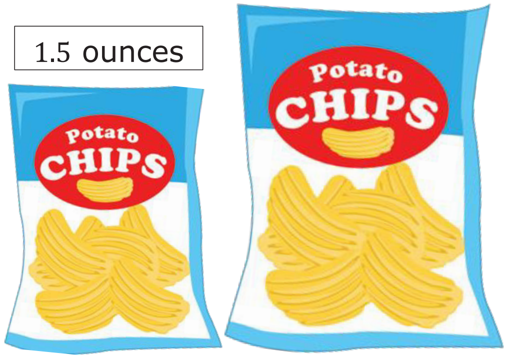
- ☐ I can use ratios and rates to describe relationships between quantities.
- ☐ I can write a ratio as a unit rate.

5.3.1 Warm-Up: Would You Rather?

1. Would you rather share the small bag of chips with 1 friend or the large bag of chips with 8 friends? Explain your reasoning using mathematics.

8 ounces

1.5 ounces



5.3.2 Collaboration: Expressing Ratios in Different Ways

Ratios are often expressed in the real world as an amount of "something per something." For example, in setting the world record, Joey Chestnut ate 7.4 hot dogs per minute.

Discuss each of the following scenarios with your partner and work together to answer the questions.

1. A recipe for marinara sauce requires 56 ounces (oz) of whole peeled tomatoes. The recipe yields 8 servings of marinara sauce.

   a. What is the ratio of tomatoes to sauce?

   b. Complete the tape diagram to represent the ratio of tomatoes to sauce.

Tomatoes (ounces)								
Sauce (servings)	1	1	1	1	1	1	1	1

   c. How many ounces of whole peeled tomatoes are in 1 serving of marinara sauce?

   d. Complete the ratio.

$$\frac{\underline{\hspace{2cm}}\text{ oz of tomatoes}}{1 \text{ serving}}$$

2. In 2012, there were 982 farms in Hardee County, Florida. The total amount of farmland was 273,978 acres.

   a. Write the ratio of acres : farms.

   b. Complete the statement.

   The total amount of farmland in Hardee County is split up among all the farms, so to determine the average number of acres for 1 farm,

   - ○ add 982 to 273,978.
   - ○ subtract 982 from 273,978.
   - ○ multiply 273,978 by 982.
   - ○ divide 273,978 by 982.

   c. Complete the ratio.

   $$\frac{\_\_\_\_\_\_\_\text{acres}}{1\text{ farm}}$$

   d. Discuss with your partner whether you think this number of acres per 1 farm means that every farm in Hardee County was that size in 2012. Then, write a brief summary of your reasoning.

5.3.3 Guided Instruction: Understanding Rates

Each of the ratios discussed so far in this lesson are also ***rates***.

> A ***rate*** is a ratio that compares two quantities of different units.

A ratio is a comparison of two numbers, so all rates can also be expressed using ratios. Rates are often used to express how long it takes to do something.

1. Consider a familiar scenario involving a rate.

   Tyler ran 2 miles in 30 minutes.

   a. Draw a tape diagram for the ratio that represents Tyler's run.

   b. Complete the ratios. Then, complete each statement to interpret the ratios.

   $$\frac{\_\_\_\_\_\_\_\text{miles}}{1\text{ minute}} \qquad \frac{\_\_\_\_\_\_\_\text{minutes}}{1\text{ mile}}$$

   Tyler runs _______ miles for every 1 minute.

   Tyler runs _______ minutes for every 1 mile.

c. Discuss with your partner how to use one of the rates to determine how many miles Tyler runs in an hour. Then, complete the statement.

Tyler runs _______ miles per hour.

Each of the rates used to describe Tyler's run is a ***unit rate***.

A ***unit rate*** is a ratio comparing a number of units of one quantity to 1 unit of a second quantity.

5.3.4 Guided Practice: Using Ratios and Rates

1. A nutritional label for raspberry ice cream is shown.

Nutrition Facts

3 servings per container

Serving size **2/3 Cup (123g)**

	Per Container	
Calories	**1000**	
		% Daily Value*
Total Fat	58g	**74%**
Saturated Fat	36g	**180%**
Trans Fat	1.5g	
Cholesterol	385mg	**128%**
Sodium	550mg	**24%**
Total Carbohydrate	106g	**39%**
Dietary Fiber	0g	**0%**
Total Sugars	88g	
Includes Added Sugars	72g	**144%**
Protein	16g	
Vitamin D	3mcg	15%
Calcium	405mg	30%
Iron	2mg	10%
Potassium	532mg	10%

*The % Daily Value tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.

a. What is the ratio of calories to servings?

b. What is the ratio of total carbohydrates in the container to the number of calories in the container?

c. What is the ratio of added sugars to the number of servings?

d. Complete the unit rate to express the number of added sugars per serving.

$$\frac{\_\_\_\_\_\_ \text{ grams}}{1 \text{ serving}}$$

5.3.5 Your Turn!

1. Tracy McGrady is a former NBA basketball player from Bartow, Florida. He played in 938 games in the NBA, scoring a total of $18{,}381$ points. Complete the ratio to express the number of points he scored per game as a unit rate. Round the number of points to the nearest tenth.

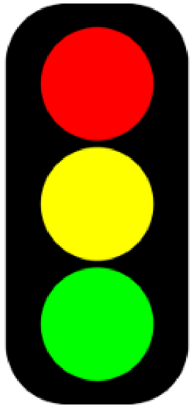
$$\frac{\_\_\_\_\_\_\text{ points}}{1\text{ game}}$$

2. The ruby-throated hummingbird is the most common hummingbird found in Florida. This bird can flap its wings $1{,}590$ times in 30 seconds. Complete the ratio to express how fast the ruby-throated hummingbird flaps its wings per second.

____ wing flaps : 1 second

5.3.6 Wrap-Up: Reflection

1. Complete the stoplight reflection to share how you are feeling about the concepts learned in today's lesson.



I still feel stuck about how to …

I'm starting to get the hang of …

I feel confident and ready to go with …

Unit 5, Lesson 4: Exploring Tables of Equivalent Ratios

Benchmarks

MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.

MA.6.AR.3.3 Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios.

Learning Target

☐ I can build tables of equivalent ratios incorporating an understanding of unit rate.

5.4.1 Warm-Up: Balloon Lift

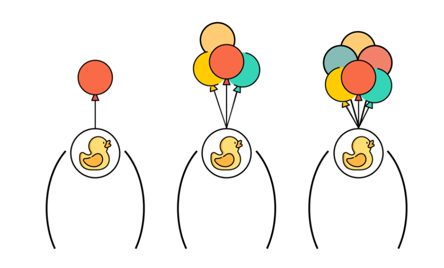
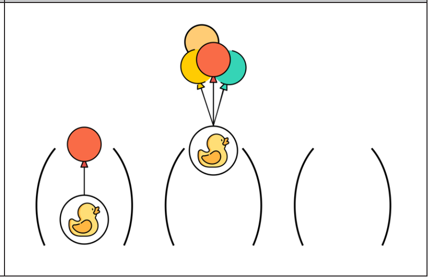
In 2015, a Canadian man attached enough balloons to his lawn chair to float up into the sky.

1. Write a question, a wonder, or a comment about this scenario to be used during a class discussion.

5.4.2 Exploration: Make It Float

Flying high with balloons is dangerous! This activity will focus on determining how many balloons it takes to float various objects.

The images below show different numbers of balloons attached to rubber ducks of equal weight. The first image shows the ducks before they are released from the hold. The second image shows what happens after they are released.

Before Release	After Release
	

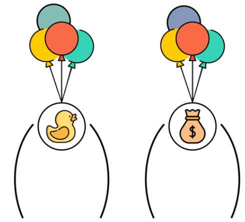
1. Write one thing you know, one thing you wonder, and one thing you learned after looking at the figures above.

   a. One thing I know is …

   b. One thing I wonder is …

   c. One thing I learned is …

2. It took 4 balloons to make the duck float. The duck weighs 56 grams. If each balloon lifts the same amount, how many grams would float with 1 balloon?

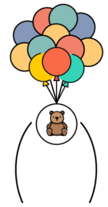


3. How many balloons will it take to float a toy bear that weighs 350 grams?

   a. Write your prediction in the table.

Object	Weight (grams)	Number of Balloons
Sandbag	14	
Duck	56	4
Bear	350	

   b. Explain the reasoning for your prediction.



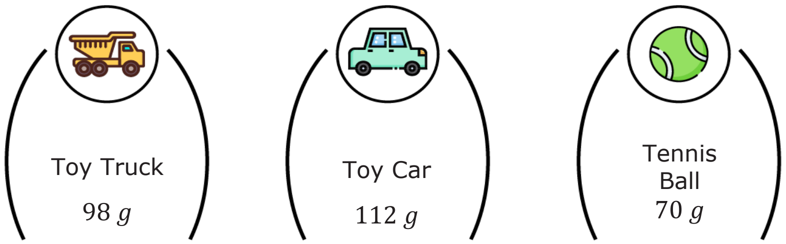
Bear

$350\ g$

4. Complete the table for the number of balloons needed to float each object based on its weight.

Object	Weight (grams)	Number of Balloons
Sandbag	14	
Duck	56	4
Bear	350	
Toy Truck	98	
Toy Car	112	
Tennis Ball	70	

5. Draw the balloons for each object that would make it float.



5.4.3 Guided Practice: Exploring Equivalent Ratios Using Tables

We have looked at the number of balloons needed to lift an object of a certain weight.

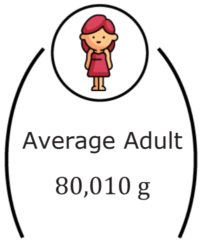
1. Use the relationship between the number of balloons, B, and the weight of the object in grams, g, to write the unit rate of $\frac{\text{weight in grams}}{\text{number of balloons}}$.
   a. Complete the table to show the unit weight lifted by a balloon.

Object	Weight (grams)	Number of Balloons	Unit Rate (grams per balloon, $\frac{g}{B}$)
Sandbag	14	1	$\frac{g}{B} = \square$
Duck	56	4	$\frac{\square}{\square} = \square$
Bear	350	25	$\frac{\square}{\square} = \square$
Toy Truck	98	7	$\frac{\square}{\square} = \square$
Toy Car	112	8	$\frac{\square}{\square} = \square$
Tennis Ball	70	5	$\frac{\square}{\square} = \square$

b. What do you notice after completing the table?

2. A diagram of an average adult whose weight is 80,010 grams (g) is shown.

   a. Discuss with your partner how to use the information from the previous table to determine how many balloons it would take to float the person.



   b. Complete the statement.

   It would take __________ balloons to float the person.

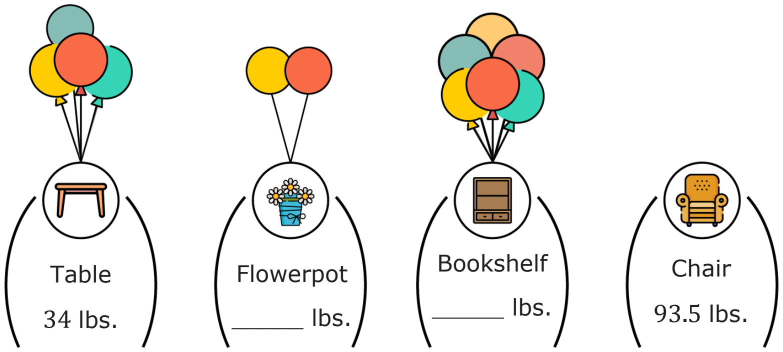
3. Weather balloons would be better to use in this situation. Unlike regular balloons, weather balloons are designed to lift heavier objects.

   The partially completed table shows the weight in pounds (lbs) of four objects and the number of weather balloons needed to float each object.

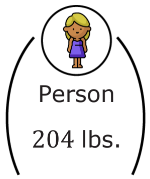
   a. Complete the table with the weight, the number of weather balloons, and unit rate for each object.

Object	Weight (lbs)	Number of Weather Balloons	Unit Rate (lbs per balloon)
Table	34	4	$\frac{\square}{\square} = \square$
Flowerpot		2	$\frac{\square}{\square} = \square$
Bookshelf		6	$\frac{\square}{\square} = \square$
Armchair	93.5		$\frac{\square}{\square} = \square$

b. Draw the weather balloons needed to make the chair float and write the weight of each object determined from the table.



4. Consider the figure shown.



How many weather balloons would it take to float the person? Write your answer in the table.

Object	Weight (lbs)	Number of Weather Balloons	Unit Rate (lbs per balloon)
Person	204		$\frac{\square}{\square} = \square$

5.4.4 Wrap-Up: Cool Down

1. Your friend wants to know how to calculate the number of balloons she would need to float.

   Answer the questions in the space provided in the table.

What information do you need to know?	
What instructions would you give her about how to calculate the number of balloons she would need to float?	

Unit 5, Lesson 5: Connecting Ratios and Rates

Benchmark

MA.6.AR.3.3 Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios.

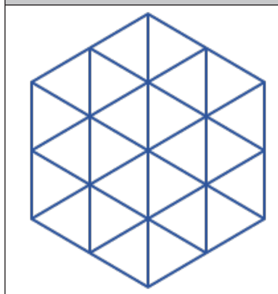
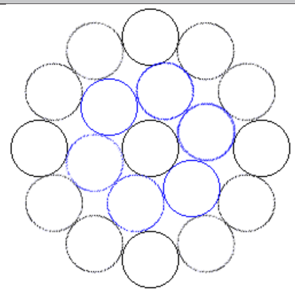
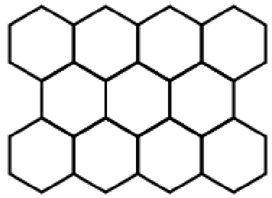
Learning Target

☐ I can make connections between ratios, rates, and unit rates.

5.5.1 Warm-Up: Why Do Honeybees Love Hexagons?

Honeybees are actually amazing mathematicians. Scientists claim they can even calculate angles. They also live inside one of the most mathematically efficient architectural designs there is: the beehive. It takes a lot of work for bees to produce wax. The hexagonal shape of the cells in a honeycomb allows bees to store the most amount of honey while using the least amount of wax. This design likely evolved over time.

Consider the shapes used in the designs shown.

Triangular Cells	Circular Cells	Hexagonal Cells
		

1. How do the designs demonstrate that the hexagonal shape allows bees to store the most amount of honey using the least amount of wax?

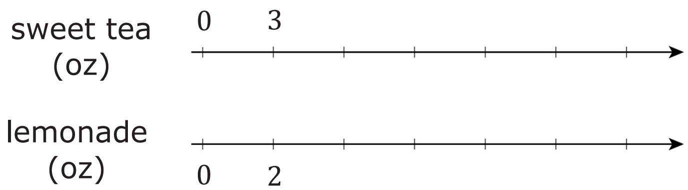
2. This is an example of mathematics in nature. What is another example of mathematics in nature you can think of?

5.5.2 Guided Instruction: Connecting Ratios and Rates

A given ratio or rate can be used to generate many equivalent ratios, which can be helpful in solving problems. Double number lines, tape diagrams, mixture charts, and ratio tables are useful tools for finding equivalent ratios.

1. Noah's favorite drink is a mixture of sweet tea and lemonade. He uses the ratio of 3 ounces (oz) of sweet tea to 2 oz of lemonade to make it.

   A double number line diagram is shown to represent the situation. Complete the double number line diagram.



A table can also be used to represent this relationship.

2. Complete the table to show how many ounces of each beverage are needed to make different-sized batches of Noah's favorite drink.

Sweet Tea (oz)	Lemonade (oz)
3	2
21	
	10
4.5	
	32
1.5	

Recipes are scaled up to make larger batches with the same taste. They can also be scaled down to make smaller batches with the same taste. Equivalent ratios can be used to determine the amount of ingredients needed when scaling a recipe up or down.

3. When scaling a recipe up, such as to find how many ounces of sweet tea to mix with 32 ounces of lemonade, which operation is used?

4. When scaling a recipe down, such as to find how many ounces of lemonade to mix with 1.5 ounces of sweet tea, which operation is used?

5. William noticed he could use similar thinking with the double number line diagram. He drew an arrow from 3 to 15 on the top number line and another arrow from 2 to 10 on the bottom number line. Complete the statement to describe what he saw.

   William noticed the ratio $\frac{3}{2}$ can be used to find the ratio $\frac{15}{10}$ by
   - ○ adding 12 to both parts of the ratio.
   - ○ multiplying both parts of the ratio by 5.
   - ○ dividing both parts of the ratio by 5.

Equivalent ratios can be found by multiplying or dividing each part of a ratio by the same value.

6. Show how the ratios $3:2$ and $48:32$ from the table are equivalent, using either multiplication or division.

7. Shay noticed the table of Noah's favorite drink included a unit rate. Circle the unit rate in the table. Then, complete the statement to explain its meaning in context.

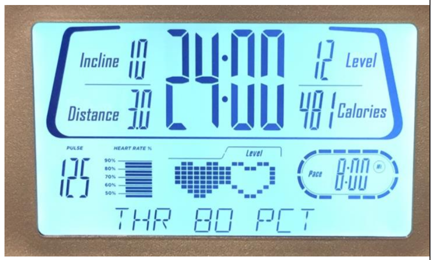
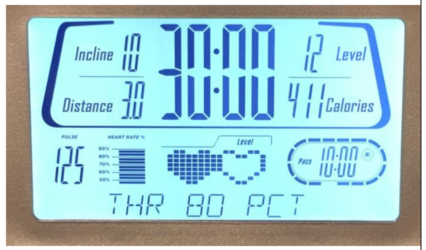
   This unit rate means there are _____ ounces of

   ____________________ in the drink for each ounce of

   ____________________.

5.5.3 Collaboration: Considering Ratios and Rates in the Context of Speed

Mai and Jada each ran on a treadmill. The treadmill display shows the distance, in miles, each person ran and the amount of time it took them, in minutes and seconds.

Mai's Treadmill Display	Jada's Treadmill Display
	

1. Discuss with your partner the similarities and differences between Mai's and Jada's workouts. Write them in the table below.

Similarities	Differences

2. Express each person's workout as a ratio of distance to time, including units.

	Ratio of Distance to Time
Mai's Workout	
Jada's Workout	

3. Each person ran at a constant rate by setting the speed on their treadmill. Discuss with your partner who ran at a faster rate using the ratios.

4. Eli thought it would be helpful to rewrite the ratios as unit rates when deciding who ran faster. His incomplete work is shown. Work with your partner to complete his work.

Mai's Workout	Jada's Workout
$\div 3$ 24 min : 3 mi $\div 3$ 8 min : 1 mi $\div 8$ ____ : ____ $\div 8$	$\div 3$ 30 min : 3 mi $\div 3$ $\div 10$ 10 min : 1 mi $\div 10$ ____ : ____

5. Complete the statements.

It took Mai _____ minutes to run 1 mile, and it took Jada _____ minutes to run 1 mile. This shows that ○ Mai ○ Jada ran faster because it took her ○ less ○ more time to run the same distance.

In 1 minute, Mai ran _____ mile(s) and Jada ran _____ mile(s). This shows that ○ Mai ○ Jada ran faster because she ran ○ less ○ more distance in the same amount of time.

5.5.4 Try It: Rates and Ratios in Context of Speed

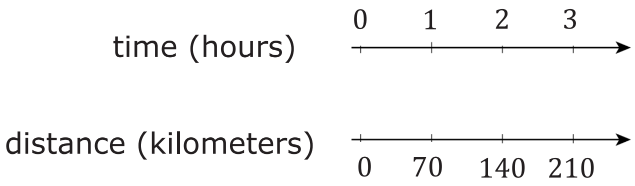
1. A slug travels 3 centimeters in 3 seconds. A snail travels 6 centimeters in 6 seconds. Ethan says, "The snail was traveling faster because it went a greater distance."

   a. Discuss Ethan's statement with your partner.

   b. Using ratios or rates, explain whether you agree with Ethan.

5.5.5 Your Turn!

1. Nadia is driving at a constant speed on a long stretch of highway on a long drive from Melbourne, Florida, to Atlanta, Georgia. Her distance traveled over time is shown on the double number line.



How far did Nadia travel at this speed for 5 hours? Explain or show your reasoning.

2. Two cleaning solutions were created to clean tile floors as shown.

Mixture A	Mixture B
5 cups of water 2 cups of vinegar	15 cups of water 8 cups of vinegar

a. Explain why the mixtures will not result in the same cleaning solution.

b. Explain which solution will have a stronger smell of vinegar.

5.5.6 Wrap-Up: Growth Mindset

Complete each statement.

1. Today I am still unsure about …

2. To fill this gap, I intend to …

Unit 5, Lesson 6: Using Two- and Three-Column Tables of Equivalent Ratios

Benchmarks

MA.6.AR.3.3 Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios.

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

Learning Targets

- ☐ I can use two- and three-column tables to display equivalent ratios.
- ☐ I can solve problems using two- and three-column tables of equivalent ratios.

5.6.1 Warm-Up: Different Ways of Expressing Division

In many Latin American countries and other countries around the world, the colon $(:)$ is used instead of $\div$ to express division. Therefore, the expression "28 divided by 4" would be expressed as $28:4$ instead of $28 \div 4$.

1. Using what you have learned about ratios, explain how $28:4$ and $28 \div 4$ can be used to express the same value.

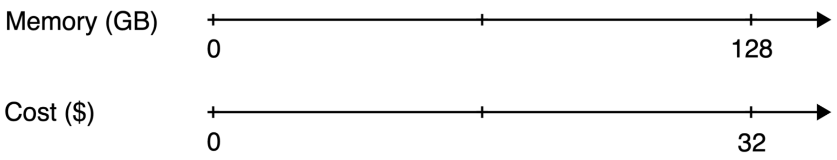
2. Write one question that comes to your mind after reading the information shown.

5.6.2 Exploration: Double Number Lines Versus Two-Column Tables

Work with your partner on the following.

In 2016, 128 gigabytes (GB) of portable computer memory cost \$32.

A double number line diagram that represents the situation is shown.



1. One set of tick marks has already been drawn to show the result of multiplying 128 and 32 each by $\frac{1}{2}$. Label the amount of memory and the cost for these tick marks.

2. Next, keep multiplying by $\frac{1}{2}$ and drawing and labeling new tick marks, until you can no longer clearly label each new tick mark with a number.

3. Discuss with your partner whether or not it's possible to clearly represent the cost of 1 GB of memory on the double number line diagram.

4. If it is possible, circle the cost of 1 GB of memory on the double line diagram. If it is not possible, explain why not.

5. A table representing the same situation is shown. Work with your partner to find the cost of 1 GB. Use as many or few rows as needed.

Memory (gigabytes)	Cost (dollars)
128	32

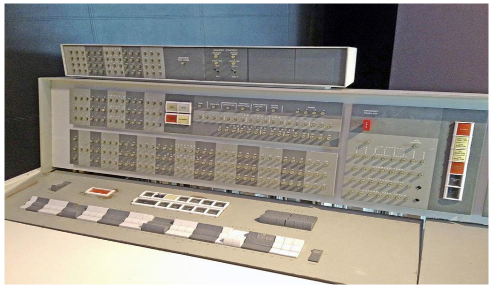
6. Explain your steps for determining the cost per gigabyte using the table.

7. Use either the double line diagram or the ratio table to determine the cost of 512 GB of portable computer memory at this rate.

8. Discuss with your partner which representation of the equivalent ratios you preferred using to find the costs of 1 GB and 512 GB. Summarize your discussion.

The IBM 7094 was regarded as one of the most powerful advanced mainframe computers in the early 1960s. NASA and the Air Force used it in the Gemini and Apollo space programs.

Despite its large size and historical impact, the IBM 7094 only had a core memory of about 150 kilobytes (KB). That's only enough to manage a few Microsoft Word documents.



Today, you can get a small USB flash drive like the one shown for less than $\$20$ that can store 4 gigabytes (GB) of memory.

That's almost 27,000 times the memory as the IBM 7094, and you can fit it in your pocket!

5.6.3 Guided Instruction: Using Three-Column Ratio Tables

Ratios can be used to make part-to-part or part-to-whole comparisons. When comparing parts of a mixture to the whole, three-column ratio tables can be useful to represent the relationships between quantities.

1. Kiara is practicing her free-throw shot at basketball practice. She averages 4 made shots for every 2 missed shots. The table shown can be used to represent this situation.

Number of Shots Made	Number of Shots Missed	Total Shots Taken
4	2	

   a. When determining this rate during practice, Kiara missed 8 free-throw shots. Complete the second row of the table to show how many shots she made and how many total shots she took.

   b. How was each value in the second row determined?

2. Kiara had the opportunity to take 3 free-throw shots in her most recent game, and her accuracy remained the same as in practice. Complete the third row of the table to show how many free-throw shots she made and missed in the game.

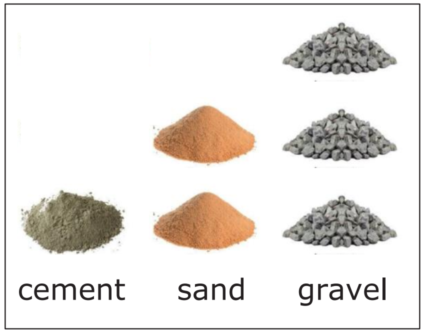
   a. How was each value in the third row determined?

   b. Complete the statement.

   For every 3 free-throw shots taken, Kiara makes _____ shot(s) and misses _____ shot(s) on average. The ratio of missed shots to made shots to total shots is _____ : _____ : _____.

3. Three-column tables can also be useful when comparing mixtures that contain more than two parts.

   Concrete can be created by mixing 1 part cement with 2 parts sand and 3 parts gravel. These dry ingredients are first combined before being mixed with water.



   a. Complete the table to determine equivalent mixtures of concrete. Show how the values in each row are determined using arrows and by indicating the factor used.

Cement	Sand	Gravel
1	2	3
		1
6.5		
	10	

   b. Marc uses one-gallon buckets to measure out the quantities needed in his concrete mixture. How many gallons of dry mix will he have in one batch of concrete?

5.6.4 Guided Practice: Using Two- and Three-Column Ratio Tables

1. A particular shade of maroon paint calls for 5 cups of red paint to be mixed with 3 cups of blue paint. Complete the table to find the quantities needed for different batches of paint.

Red Paint (cups)	Blue Paint (cups)	Maroon Paint (cups)
5	3	
	12	
		2
2.5		
		80

a. If the mixture in the first row represents one batch of maroon paint, how many batches are in 80 cups of maroon paint?

b. Explain how this relates to the factor used to complete the last row in the table.

2. There are 16 cups in one gallon. Use this information and the table below to determine how many gallons of paint the 80-cup mixture of maroon paint makes.

Cups	Gallons

3. In April, Kyla went for a walk on 4 days for every 1 day that she did not walk. There are 30 days in April. Use this information to complete the ratio table.

Walking Days	4	
Non-Walking Days	1	
Total Days		30

Complete the statement.

Throughout the year, Kyla continued walking 4 days for every 1 day that she did not walk. Based on this, in November, Kyla walked_____ of the 30 days.

5.6.5 Wrap-Up: Ticket Out the Door

1. Julio participated in a walk-a-thon to raise money for cancer research. Walking at a constant speed, Julio traveled 6.4 miles in the first 2 hours of the walk-a-thon. His goal is to walk for 7 hours. Use the table to find how far Julio will have walked if he reaches his 7-hour goal. Show how the values in each row are determined.

Hours	Miles Walked
2	6.4

Unit 5, Lesson 7: Digging Deeper Into Unit

Benchmarks

MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

Learning Targets

- ☐ I can determine a unit rate when given different ratios.
- ☐ I can interpret unit rates in real-world contexts.
- ☐ I can use unit rates to solve problems.

5.7.1 Warm-Up: Ticket Booth

At a local fall festival, the following prices are shown at the ticket booth.

Ticket Prices	
1 Ticket	$0.50
12 Tickets	$5.00
25 Tickets	$10.00
50 Tickets	$25.00
120 Tickets	$50.00

1. Without performing any calculations, explain which option you think would be the best.

2. Determine a pair of ratios, tickets:price, that appear to be equivalent. Write them in the space below.

   If it is not possible to determine a pair of equivalent ratios from the sign, select one ratio and write another one that is equivalent.

   ____ ticket(s) : $_______

   ____ ticket(s) : $_______

3. By what factor can both values in the first ratio be multiplied to find the second ratio?

5.7.2 Guided Instruction: Interpreting Unit Rate

A ***unit rate*** is a ratio comparing a number of units of a quantity to ____ unit of another quantity.

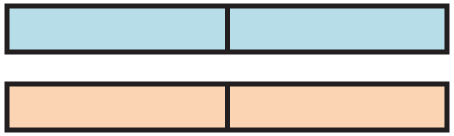
1. Several rates are listed. Circle those that represent a unit rate.

earning $42 for 3 hours	35 miles per gallon
75 miles per hour	2.5 miles in 24 minutes
each banana costs $0.29	12 inches for every 1 foot

2. Tia was hiking on the Florida National Scenic Trail. She hiked a total of 4.5 miles in 2 hours. The tape diagram models the distance and time she hiked.

Tia hiked 4.5 miles.

Tia hiked for 2 hours.



Notice that each tape diagram is the same size because it represents quantities in the same situation.

a. Fill in each box of the tape diagram to show the value it represents.

b. What is Tia's unit rate in miles per hour?

c. Draw a tape diagram that represents the time it would take Tia to hike 9 miles on the trail at this pace.

3. Carter bikes $12\frac{1}{2}$ miles in 5 hours around the city.

a. Work with your partner to fill in each box of the tape diagram to show the value it represents. Be sure to include units.

b. Interpret the unit rate for Carter's bike ride. Include units.

c. Explain which person is moving slower, Tia or Carter. Include the unit rates in your explanation.

4. Complete the statement.

In each scenario, the unit rate in [○ hours per mile / ○ miles per hour] can be determined by [○ dividing / ○ multiplying] the number of [○ hours / ○ miles] traveled by the number of [○ hours / ○ miles] traveled.

5.7.3 Guided Practice: Using Unit Rates to Solve Problems

Unit rates would be helpful in considering the ticket booth from the Warm-Up. Miguel set up the table shown to consider the costs per ticket for each advertised price.

1. Work with your partner to complete the table. Then, answer the questions that follow.

Total Cost	Number of Tickets	Cost per Ticket
\$0.50	1	\$0.50
\$5.00	12	
\$10.00	25	
\$25.00	50	
\$50.00	120	

a. Using the unit rates, explain which ticket option(s) is(are) the best deal.

b. Using the unit rates, explain which ticket options are the same deal.

c. Compare your answer to the previous question to the equivalent ratios you determined in the Warm-Up. Make any revisions to your previous thinking, as necessary.

d. With your partner, discuss recommendations for how the high school should change its prices for carnival tickets.

2. Holly, Frank, and Ella work part-time for Shoes-R-Us. The table shows how much money each person earns during a work week.

Employee	Weekly Pay	Number of Hours Worked
Holly	\$165	20
Frank	\$162	18
Ella	\$170	20

Work with your partner to answer the following questions using the information in the table.

a. Find and interpret the hourly rate Holly earns.

b. Which employee earns the higher hourly rate, Ella or Frank?

c. If Frank works 25 hours next week, how much should he expect to earn?

5.7.4 Collaboration: Using and Interpreting Unit Rates

Work with your partner to interpret the unit rates for each situation.

1. Tyler paid $\$12$ for 4 large candy bars.

   a. Tyler interprets the unit rate as $\frac{\$3}{1 \text{ bar}}$.
   His friend Ben interprets the unit rate as $\frac{1 \text{ bar}}{\$3}$. Which person interpreted the unit rate correctly?

   b. Based on the unit rate, how many candy bars can Tyler purchase for $\$45$?

2. Aaliyah's family drinks a total of 10 gallons of milk every 6 weeks and 4 gallons of apple juice every 2 weeks.

   a. Which beverage does Aaliyah's family drink more of in one week?

   b. How many weeks does it take the family to consume 1 gallon of milk?

3. Carlos buys 24 bottles of spring water for $\$3.60$ and 3 jars of pasta sauce for $\$9.69$.

   a. What does each item, a bottle of spring water and a jar of pasta sauce, cost per unit?

   b. What would be Carlos' total cost if he purchased 10 bottles of spring water and 2 jars of pasta sauce at their unit prices?

5.7.5 Bring It Together: Interpreting Unit Rates

Work with your partner to write three statements to summarize your learning about unit rates from today's lesson.

1.

2.

3.

5.7.6 Wrap-Up: Ticket Out the Door

A cheesecake recipe says, "Mix 12 ounces (oz) of cream cheese with 15 oz of sugar."

1. How many ounces of cream cheese are needed per ounce of sugar?

2. How many ounces of sugar are used for each ounce of cream cheese?

3. Max plans to make more than one cheesecake. How much cream cheese will she need to mix with 35 oz of sugar?

Unit 5, Lesson 8: Applying Ratio Relationships to Percentages

Benchmark

MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.

Learning Targets

- ☐ I can make connections between ratios and percentages.
- ☐ I can use double-line diagrams, tape diagrams, and ratio tables to solve problems involving percentages.

5.8.1 Warm-Up: Estimate the Percentage

A photo of blue and pink thumb tacks is shown.

1. Based on what is shown in the picture, predict the percentage of thumb tacks that are blue by completing each statement.

   I know for sure that more than ______% of the thumb tacks are blue.

   I know for sure that less than ______% of the thumb tacks are blue.

2. My best estimate is that ______% of the thumb tacks are blue.

5.8.2 Refresher: Percentages

A ***percentage*** is a ratio expressed as a fraction of 100.

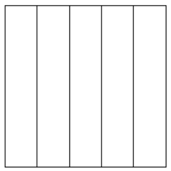
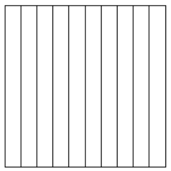
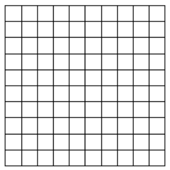
> ***Percent*** $(\%)$ means "per 100," "for every 100," or "out of 100."

Work with your partner to complete the following.

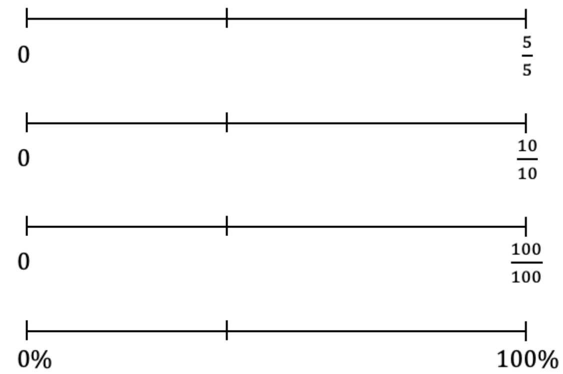
1. Each of the large squares shown are the same size.

   a. Shade $\frac{2}{5}$ of each model.

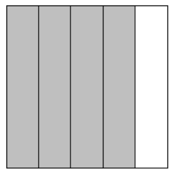
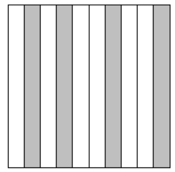
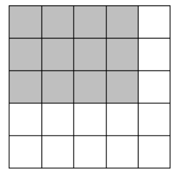
   b. Then, express the shaded part of each model as a fraction and as a percentage.

Model			
Fraction			
Percentage			

   c. Label the identified tick marks on each number line below to match the models.



2. Express the shaded part of each model as a fraction and as a percentage.

Model			
Fraction			
Percentage			

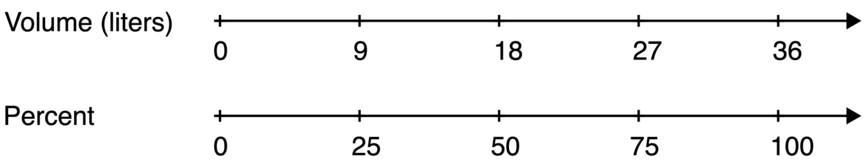
3. Complete the statements.

   Percentages represent ○ part-to-part ○ part-to-whole relationships where each percentage is a part per 100. In this case, 100% is ○ only part of the total. ○ the total, or whole, amount.

5.8.3 Guided Instruction: Using Ratio Models to Represent Percentages

Double number lines, tape diagrams, and ratio tables can be used to model percentages similarly to other ratios. These can be helpful for determining equivalent ratios and solving problems involving percentages.

1. Consider a cooking pot that can hold 36 liters of water. The double number line diagram models this situation.



a. What percentage of the pot is filled when it contains 9 liters of water?

b. Circle where the answer to the previous question is represented on the number line diagram.

c. How many parts were the totals on each number line, 36 liters and 100%, split into?

d. An equivalent ratio table can also be used to model similar thinking. Determine the factor that was multiplied by 36 to get 9, and use it to complete the blanks below.

× ______

Volume (liters)	Percentage
36	100
9	

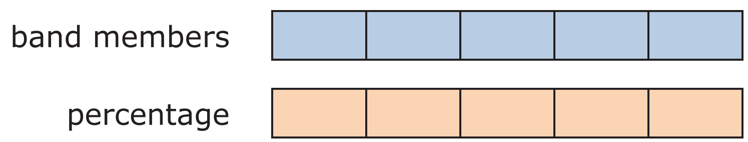
× ______

e. Describe how the factor used to find the equivalent ratio in the table relates to how the number lines were divided in the number line diagram.

Tape diagrams can also be used to represent problems involving percentages.

2. At Millennium Middle School, there are 70 students in the school band. Of the band members, 40% are sixth graders, 20% are seventh graders, and the rest are eighth graders.

The tape diagrams shown represent the situation.



a. Label each small rectangle in the tape diagram of the band members with the number of students it represents.

b. Label each small rectangle in the tape diagram of the percentage of band members with the value it represents.

c. How many band members are sixth graders?

d. Circle the part of the tape diagram that represents these students.

e. How many band members are seventh graders?

f. Draw a rectangle around the part of the diagram that represents these students.

g. What percentage of the band members are eighth graders?

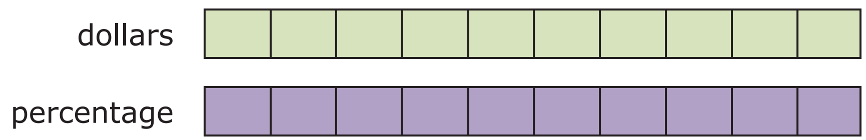
h. How many band members are eighth graders?

5.8.4 Guided Practice: Using Ratio Models to Represent Percentages

1. Use the number line diagram to determine what percentage of 80 is 24. Then, complete the statement.

24 is _____% of 80.

2. Use the tape diagram to determine 70% of $\$60$. Then, complete the statement.



70% of $\$60$ is $_____.

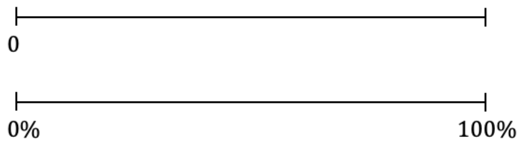
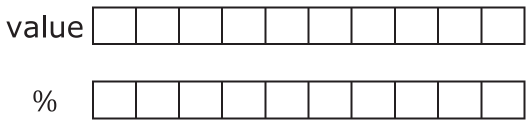
3. During a sale, a pair of jeans is on sale for 80% of its regular price. The regular price of the jeans is $\$75$. Complete the ratio table to determine the sale price. Then, complete the statement.

Price ($)	Percentage
	80
	100

80% of $\$75$ is $_____.

5.8.5 Your Turn!

1. Use the models to find the values that complete each statement. Show your work.
   For the last row, use the model of your choice.

<table>
<tr><th>Statement</th><th>Model</th></tr>
<tr><td>20% of 60 is _____.</td><td>
</td></tr>
<tr><td>_____% of 25 is 12.</td><td><table><tr><th>Value</th><th>Percentage</th></tr><tr><td></td><td>100</td></tr><tr><td></td><td></td></tr></table></td></tr>
<tr><td>40% of _____ is 28.</td><td>
</td></tr>
<tr><td>16% of 125 is _____.</td><td></td></tr>
</table>

5.8.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is related to each target.

1. I can make connections between ratios and percentages.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

2. I can use double-line diagrams, tape diagrams, and ratio tables to solve problems involving percentages.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

3. Write one question you still have about solving problems involving percentages after today's lesson.

Unit 5, Lesson 9: Applying Ratios to Solve Problems Involving Percentages

Benchmark

MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.

Learning Target

- ☐ I can apply ratio relationships to solve problems involving percentages.

5.9.1 Warm-Up: Percentages on the Number Line

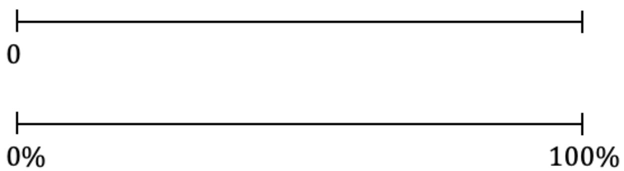
1. Place a digit in each blue box to create an accurate number line. Use only digits 0 to 9, at most one time each. The number line is not drawn to scale.

5.9.2 Guided Practice: Applying Ratio Relationships to Solve Percentage Problems

1. There is a 10% off sale on laptop computers. Ryan saved $\$35$ off of the original laptop price during the sale. Use the ratio table to determine the original price of the laptop.

Dollar Amount	Percentage
	10
	100

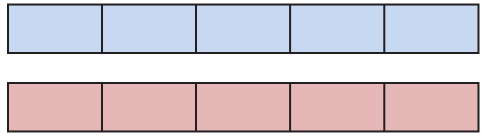
2. Lily was running in her first 10K race in Islamorada, Florida. She asked her sister to cheer for her when she was 75% of the way finished with the race so that it would motivate her to finish strong. The race is approximately 6.2 miles long. Use the number line diagram to determine how many miles Lily will have run when she is 75% finished with the race.



3. During the first part of his hike along the Myakka Hiking Trail in Florida State Park, Andre drank 1.5 liters of water. If this is 40% of the water he brought on the hike, use the tape diagrams to determine how much water he brought.

water (liters)

percentage



Andre brought ______ liters of water with him on the hike.

5.9.3 Exploration: Using Tables and Unit Rates with Percentages

A restaurant has a sign by the front door that says, "Maximum Occupancy: 75 people."

Damian completed the first two rows of a ratio table representing this situation. His work is shown.

Number of People	Percentage of Occupancy
75	100
1	$\frac{100}{75} = \frac{4}{3}$

$\times \frac{1}{75}$ $\qquad$ $\times \frac{1}{75}$

1. Work with your partner to complete the statements to describe Damian's work.

First, Damian interpreted the restaurant sign to mean that ______% is equal to 75 people.

Next, he multiplied both values in the ratio by $\frac{1}{75}$ to find the unit rate of the
- ○ percentage occupancy per 1 person
- ○ number of people per 1% occupancy

in the restaurant. He multiplied this factor by both values to find an equivalent ratio.

He then rewrote the result, $\frac{100}{75}$, as an equivalent fraction, $\frac{4}{3}$, by dividing both the numerator and denominator by _____.

2. With your partner, complete the table to find additional equivalent ratios. Use arrows and multiplication to show how you determine the values in each row.

Number of People	Percentage of Occupancy
75	100
1	$\frac{4}{3}$
9	
51	
	1
	40

3. Complete the statements to interpret the ratios.

When there are 9 people in the restaurant, it is _____% full.

The restaurant is at _____% capacity when 51 people are inside.

The restaurant is 40% full when _____ people are there.

4. Christian wondered if it is possible for the restaurant to ever be at exactly 1% capacity. Discuss his thinking with your partner. Then, write a brief summary of your discussion.

Last Sunday, $1{,}575$ people visited an amusement park.

- 56% of the visitors were adults
- 16% of the visitors were teenagers
- 28% of the visitors were children under the age of 12

5. Work with your partner to use the equivalent ratio table below to determine the number of adults, teenagers, and children who visited the park last Sunday. Show your work.

Number of Visitors	Percentage
	100

6. Complete the statements.

Last Sunday, ______ adults, ______ teenagers, and ______ children under the age of 12 visited the amusement park.

5.9.4 Your Turn!

1. At a hardware store, a tool set costs $\$80$. Using a coupon, Denise is able to pay $\$12$ less than the original price.

   a. Complete the table.

Dollar Amount	Percentage

   b. The coupon gives Denise ______% off of the original price.

2. A bathtub can hold 80 gallons of water. Isaiah fills it up to 65% capacity to take a bath.

   a. Complete the table to determine how many gallons of water Isaiah had in the bathtub.

Number of Gallons	Percentage of Capacity

   b. Isaiah filled the tub with ______ gallons of water to be at 65% capacity.

5.9.5 Wrap-Up: Ticket Out the Door

1. A large school bus can hold a maximum of 72 children. This morning, Malachi's school bus only had 54 students on board. What percentage of Malachi's school bus is full?

Number of Students	Percentage of Capacity

Unit 5, Lesson 10: Converting Units Using Ratios

Benchmark

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

Learning Target

☐ I can use ratios to convert units within the same measurement system.

5.10.1 Warm-Up: Furniture Designer

1. Jared Rusten is a woodworker who creates modern furniture in San Francisco.

   a. Jared usually starts designing furniture by sketching his ideas before he makes a scaled drawing using measurements to build the actual piece of furniture. What do you think a scale drawing is?

   b. Jared said, "I never thought that I'd be using math as much as I do now." What are some of the ways he uses math as a furniture designer and maker?

5.10.2 Guided Instruction: Converting Units of Measure Using Ratio Relationships

Many different units are used to measure lengths, weight, capacity (volume), and time.

1. What are some units of measure that are used to take each type of measurement?

Length	Weight	Capacity	Time

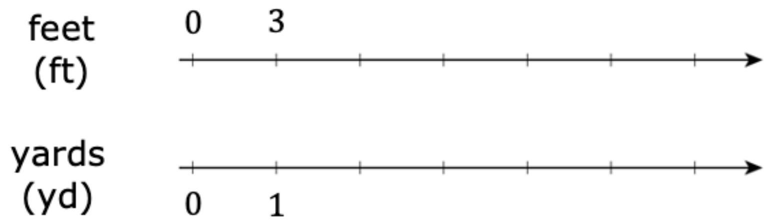
The United States uses the customary system of measurement. The table shows the unit conversions for units within this system.

2. Label each row using the words **Length, Weight,** and **Capacity** to indicate what each set of units measures.

Type of Measurement	Unit Conversions in the Customary System
	1 foot (ft) = 12 inches (in.) 1 yard (yd) = 3 feet (ft) 1 mile (mi) = 5,280 feet (ft) 1 mile (mi) = 1,760 yards (yd)
	1 cup (c) = 8 fluid ounces (fl oz) 1 pint (pt) = 2 cups (c) 1 quart (qt) = 2 pints (pt) 1 gallon (gal.) = 4 quarts (qt)
	1 pound (lb) = 16 ounces (oz) 1 ton (t) = 2,000 pounds (lb)

Each conversion rate represents a ratio. Double number line diagrams can be used to convert units.

3. Use the double number line diagram shown to find how many feet (ft) are in 4 yards (yd).



There are _____ feet in 4 yards.

Tables can also be used to convert units of measure.

4. The average weight of an African forest elephant is $6{,}000$ pounds. Use the unit conversion and the table to determine how many tons the average African forest elephant weighs.

Pounds	Tons
2,000	1

Notice that each conversion rate is a unit rate.

Unit rates can be used as factors to convert from one unit to another.

For example, the factor $\frac{4 \text{ quarts}}{1 \text{ gallon}}$ is equivalent to 1 because the numerator and denominator are equivalent values.

Multiplying by factors equivalent to 1 will result in equivalent values.

Kaluba determined how many quarts are in 2 gallons of milk using this strategy. His work is shown.

$$\frac{2 \text{ gal.}}{1} \times \frac{4 \text{ qt}}{1 \text{ gal.}} = 8 \text{ qt}$$

Kaluba explained, "In my calculation, the unit 'gallons' is in the numerator and the denominator. This means I can eliminate this unit because it becomes 1, similar to when equivalent values are in the numerator and denominator. The result of the conversion is then the number of quarts."

5. Complete the calculation to find how many fluid ounces (fl oz) are in 3 pints (pt).

$$\frac{3 \text{ pt}}{1} \times \frac{\underline{\qquad}}{1 \text{ pt}} \times \frac{\underline{\qquad} \text{ fl oz}}{\underline{\qquad}} = \underline{\qquad} \text{ fl oz}$$

> The units of measure used in the United States are ***customary units***.

> The units of measure used in most of the world are ***metric units***. Like the decimal system, the metric system uses the base 10.

The tables below show additional unit conversion rates for other units of measure.

Type of Measurement	Unit Conversions in the Metric System
Length	1 meter (m) = 100 centimeters (cm) 1 meter (m) = 1000 millimeters (mm) 1 kilometer (km) = 1000 meters (m)
Capacity	1 liter (L) = 1000 milliliters (mL)
Weight or Mass	1 gram (g) = 1000 milligrams (mg) 1 kilogram (kg) = 1000 grams (g)

Time Conversions
1 minute (min) = 60 seconds (s) 1 hour (h) = 60 minutes (min) 1 day = 24 hours (h) 1 year = 365 days 1 year = 52 weeks 1 week = 7 days

5.10.3 Guided Practice: Converting Units of Measure Using Unit Rates

1. Use the conversion rates to find each value. Show your work.

   a. 2.62 km = _____ m

   b. 9 in. = _____ yd

   c. 1 week = _____ hours

2. Adrian weighed 7 pounds (lb) and 4 ounces (oz) when he was born. How many ounces did Adrian weigh when he was born?

3. Ibuprofen is a medicine sometimes used for pain relief, such as for a headache. For adults, it is not safe to take more than 3200 milligrams (mg) of ibuprofen each day. How many grams of ibuprofen is safe for adults to take in one day?

5.10.4 Collaboration: Error Analysis

1. Valerie makes sure she drinks 8 cups (c) of water each day. She wondered how many gallons of water this is, so she used conversion rates to find out. Her work is shown.

$$\frac{80\text{ c}}{1} \times \frac{2}{1} \times \frac{1}{2} \times \frac{1}{4} = \frac{160}{8} = 20\text{ gal.}$$

   Valerie knew her answer didn't make sense because there is no way she drinks 20 gallons of water each day. She is not sure where she went wrong.

   a. Discuss Valerie's work with your partner. Then, work together to find the correct number of gallons she drinks each day.

b. Help Valerie understand and learn from her mistakes. Write a brief note to Valerie explaining what her mistakes were and how she can learn from them. Include strategies that may be helpful.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

c. Read your partner's note to Valerie and offer suggestions on ways to improve their note.

5.10.5 Your Turn!

1. Use the conversion rates to find each value. Show your work.

   a. $11{,}200$ milliliters = _____ liters

   b. 3.5 gallons = _____ cups

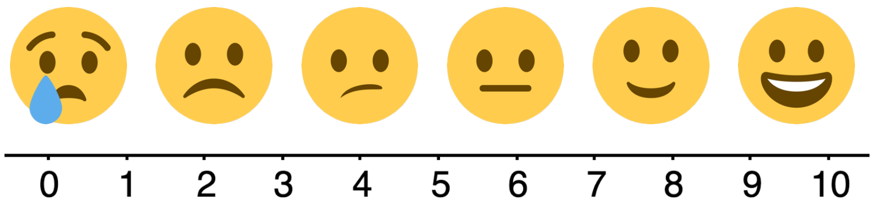
   c. 2 days = _____ minutes

2. Malik plays football for Jones High School in Orlando, Florida. Last Friday, he ran 94 yards for a touchdown on a kickoff return. How many feet did Malik run on the kickoff return?

3. About 727 pints of milk are distributed each day from the cafeteria at Fort King Middle School in Marion County, Florida. How many gallons of milk are distributed at Fort King Middle School each day?

5.10.6 Wrap-Up: Reflection

1. Use the emoji scale to indicate your level of confidence with using ratios to convert units within the same measurement system.



2. Explain why you indicated the confidence level above.

Unit 5, Lesson 11: Solving Problems Involving Ratios

Benchmark

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

Learning Target

☐ I can solve problems involving ratios, rates, and unit rates.

5.11.1 Warm-Up: Bell Work

1. Mai has to fill and seal 60 packages of homemade trail mix to sell at the upcoming farmer's market. She filled and sealed 25 packages in 32 minutes.
   Mai wonders, "If I keep working at this rate, will I be able to get all of these done in less than 1 hour?"

   Explain whether Mai will be able to get all 60 packages filled in less than 1 hour working at this rate.

5.11.2 Activity: Station 1: Is It a Deal?

Your teacher will give you a set of cards showing different offers.

1. Find Card A and work with your partner to decide whether the offer on Card A is a good deal. Explain or show your reasoning in the space provided.

2. Next, split cards B–E so you and your partner each have two cards.

   a. Decide individually if your two cards are good deals. Explain your reasoning for your two cards in the table.

Is the new deal a good deal?			
Card B	Card C	Card D	Card E

   b. For each of your cards, explain to your partner if you think it is a good deal and why.

   c. Listen to your partner's explanations for their cards. If you disagree, explain your thinking. Write your partner's explanation for their cards in the table above, after resolving any differences.

   d. Revise any decisions about your cards based on the feedback from your partner.

3. When you and your partner are in agreement about cards B–E, place all the cards you think are a good deal in one stack, and all the cards you think are a bad deal in another stack.

4. Another potential deal is shown.

Fruit Snacks

Original: 12 for $9.12
New Deal: 9 for _______

   a. What is the price for 9 packs of fruit snacks at the same unit price as the original?

   b. Complete the statement.

   Any price less than $\$$_______ for 9 packs of fruit snacks would be considered a good deal because the cost per pack would be less than $\$$_______.

5.11.3 Activity: Station 2: Cooking with a Tablespoon

Diego is trying to follow a recipe, but he cannot find any measuring cups! He only has a tablespoon.

In the cookbook, it says that 1 cup equals 16 tablespoons.

1. How could Diego use the tablespoon to measure out these ingredients?

    a. $\frac{1}{2}$ cup almonds

    b. $1\frac{1}{4}$ cup of oatmeal

    c. $2\frac{3}{4}$ cup of flour

2. Diego also adds the following ingredients. How many cups of each did he use?

    a. 28 tablespoons of sugar

    b. 6 tablespoons of cocoa powder

5.11.4 Activity: Station 3: Turbo Texting

1. Jonah was with his friend Erin after school and he needed to text his mom. Erin watched Jonah text and commented on how slow he was.

    They decided to have a text race to determine who was faster. The image below shows what each person texted in the race.

    a. Discuss any initial observations with your partner.

b. Erin was able to send her text slightly faster than Jonah. Explain if you think this means she is the faster texter.

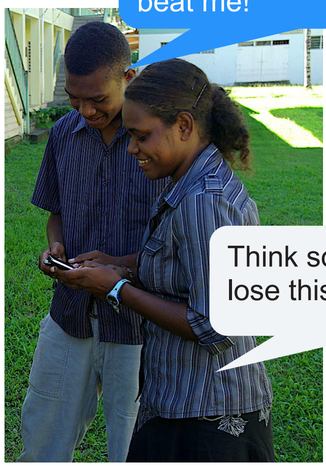
c. Describe the relationship you notice between the number of characters in a text versus how long it might take to type it.

d. What additional information would be helpful to determine who is the faster texter?

2. The image below gives additional information about each person's text.

I am the fastest texter!!! You are so slow! You will never beat me!

67 characters
7.4 seconds



Think so? You're about to lose this! It's on!!

46 characters
6.5 seconds

Work with your partner or small group to determine who is the faster texter, using your preferred strategy. Show your work.

3. Jonah and Erin decided to send one last text to see who could send it out faster. The image shows the text they each sent.

So let's settle this once and for all. I am a way faster texter and there is nothing you can do about it. I am the GOAT and you'll never catch me! I won!!

So let's settle this once and for all. I am a way faster texter and there is nothing you can do about it. I am the GOAT and you'll never catch me! I won!!

154 characters

If both people sent the message above at the same rate as before, predict about how many seconds it took each person to text the message to the nearest tenth of a second. Show your work.

Person	Predicted Time
Jonah	
Erin	

5.11.5 Wrap-Up: Reflection

1. Complete the learning period based on today's lesson.

One question I would like to have answered …

One mistake I learned from today …	One thing I am not sure about … YET!

One thing I am confident I know is …

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